

Dominik Ritz

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TECHNICAL SKILLS

Languages: Python, C, C++, Typescript, Javascript, Java, C#, Bash, Scala
Fullstack: React, Next.js, Node.js, NestJS, Express, Fastify, Flask, PostgreSQL, MongoDB, TailwindCSS, Git
Infra: AWS, Kubernetes, Docker, Terraform, GCP, DataDog, Kafka, Temporal
AI/ML: PyTorch, LangChain, pandas, NumPy, Matplotlib, spaCy

EDUCATION

University of Waterloo Waterloo, Ontario
BSE, Honours Software Engineering 2023 – 2028

EXPERIENCE

BitGo – Software Engineer Intern (Post Trade) May – Aug 2026
Go, AWS RDS/DMS, Grafana, Kafka, TypeScript New York, New York

- Migrated **5M+ rows** on 24/7 live trading platform, replacing cross-database writes with atomic transactions
- Automated token onboarding through centralized asset registry, eliminating manual configuration
- Eliminated Kafka consumer-group rebalance loops with Kubernetes health probes; improved on-call alerts

Super.com – Software Engineer Intern (Infra + DevOps) Sep – Dec 2025
Kubernetes, AWS, DataDog, Terraform Remote

- Deployed **GDPR-compliant service** on AWS (ElastiCache/RDS/EC2), **enabling EU market expansion**
- Built **AI coding agent** responsible for shipping **37 production PRs (7%)**, along with Datadog usage dashboard
- Built **AI-powered CI bot** to triage and diagnose E2E test failures, **cutting average deploy time by 14%**

Hamming AI (YC S24) – Software Engineer Intern Jan – Apr 2025
NextJS, PostgreSQL, Temporal, LiveKit, tRPC, DataDog San Francisco, California

- Built dynamically generated state machines (IVR) for **AI voice agent testing**; directly led to new customers
- Investigated and fixed critical concurrency issues with Redis that **reduced call error rate by 85%**
- **Automated frontend and backend tests** using GitHub Actions and cron jobs to catch regressions in CI/CD
- Enabled **third-party voice agents** to integrate with the Hamming platform; directly led to new customers

Carnegie Mellon University Biometrics Center – Software Engineer Intern May – Aug 2024
ROS2, C++, Python, Javascript, Docker, React, Flask Pittsburgh, Pennsylvania

- Built **autonomous grocery-delivery robot** achieving centimeter-level navigation across CMU's campus
- Built **AI checkout system** using **OpenCV** to prevent retail theft by classifying 250K+ Walmart products

WATonomous – Autonomous Software Engineer Feb – Oct 2024
ROS2, Python, C++, Docker, PyTorch Waterloo, Ontario

- Trained **graph-based trajectory prediction** model to enhance vehicle's autonomous navigation
- Reduced average displacement error by 18% and final displacement error by 22%

Palturai – Software Engineer Intern Jun – Aug 2021/2022
Java, Apache Jena, JavaFX, JSoup, Gson Paoli, Pennsylvania

- Developed an **interactive knowledge graph visualization tool** and a custom language for tailored filtering, aesthetic control, and enhanced data display (RDF-Viewer)
- **Scraped SEC** filings, extracting relational data to serve as test cases for RDF-Viewer
- **Designed Java SDK** to abstract interaction with fraud detection graph database, handling 210M+ relationships

PROJECTS

Cookd  | *Next.js, React, NestJS, Prisma, PostgreSQL, Supabase, Docker, Railway, LangChain/Groq*

- Built & launched **social cooking app**, scaling to hundreds of user posts
- Engineered **AI recipe import pipeline** that extracts structured recipes from websites and social media
- Owned infrastructure across Docker/Railway, Supabase auth/storage, Prisma migrations, and CI

Blindseer  | *Python, LLaVa, GCP, Docker, Flask, Raspberry Pi*

- Codeveloped an app leveraging **generative AI** to help visually impaired individuals navigate their environment
- Implemented a Flask server in Docker with LLaVa to **prompt image embeddings** using Vicuna 7B